DQ Decoupling Guide

Power Application Controller™

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1 DESCRIPTION

For a PMSM motor, the voltage equations based on the DQ frame of reference are:

$$V_d = R_s i_d + L_d p i_d - \omega L_q i_q$$

$$V_q = R_s i_q + L_q p i_q + \omega L_d i_d + \omega \Phi$$

The d-axis and q-axis cross-coupling terms $-\omega L_q i_q$ and $+ \omega L_d i_d$ respectively are often ignored in the current control loops. Doing so may be inconsequential in many circumstances. But there are some applications where the current control loop requires a very dynamic and fast response. In such cases, compensating for these terms reduces the dependence on the bandwidth of the PI controller to respond to these fast changes. This compensation in the current control system can be accomplished by using a DQ-decoupling technique. The goal here is to make the controllers for each axis independent from each other, so that they don't have to respond to changes in each other's axis and just focus on responding to changes in their own axis.

Additionally, the BEMF term ($\omega\Phi$) can also be compensated for and by doing so the Q-axis current controller would entirely focus on the dynamic changes.

2 FIRMWARE FLOW

The FOC firmware has two options to choose for the DQ decoupling implementation. The two options are:

1) Feedforward decoupling
2) Feedback decoupling

The basic implementation for these decoupling methods is illustrated in Figure 1 and Figure 2.

If Feedback decoupling is selected, a low-pass filter is applied to the feedback current in order to avoid issues resulting from the injection of high frequency elements into the current control loop.

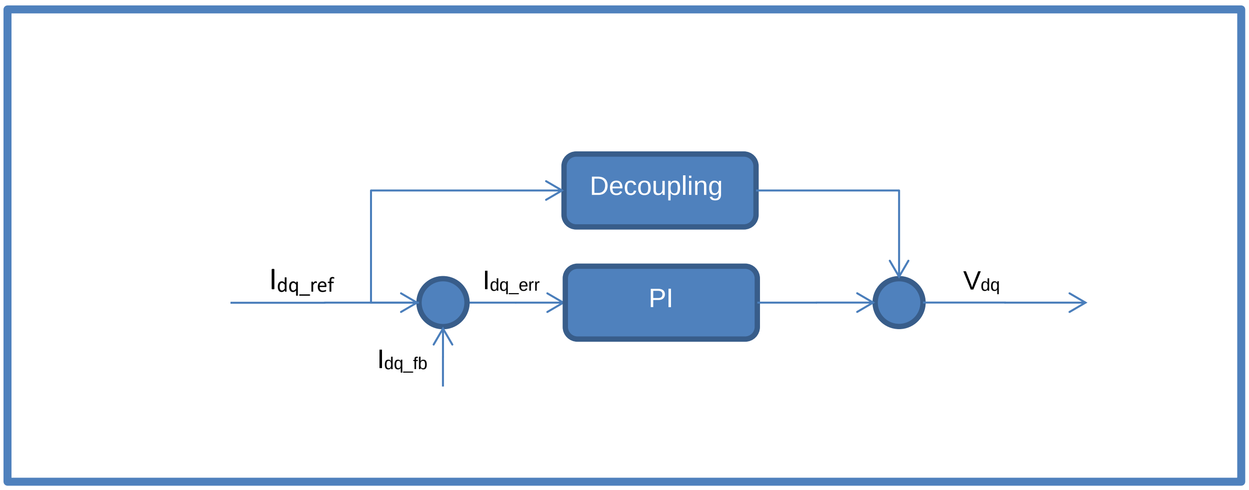

Figure 1: Feedforward Decoupling

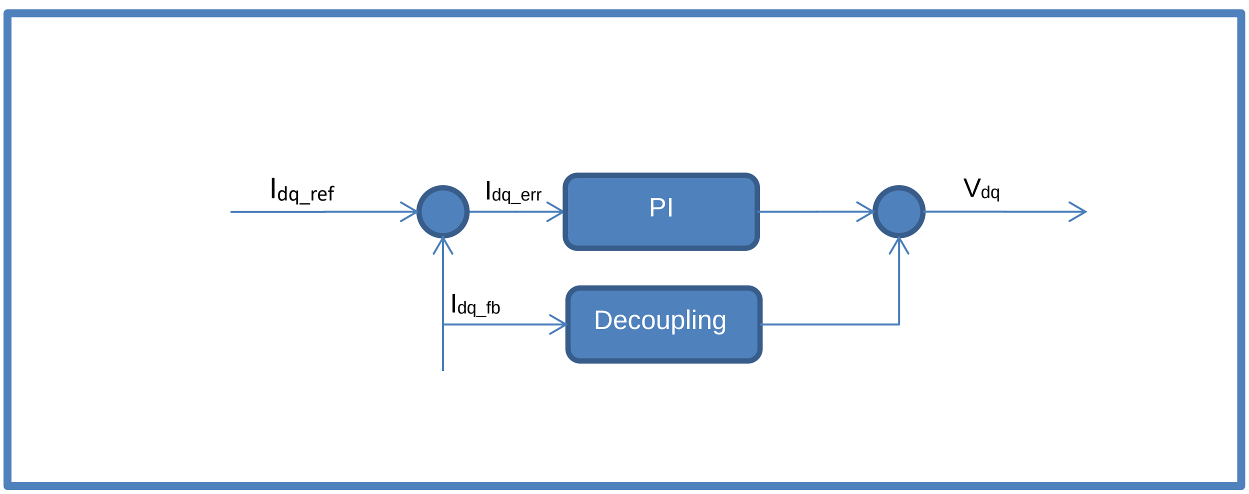

Figure 2: Feedback Decoupling

3 USING DQ DECOUPLING

3.1 Enabling DQ Decoupling

To enable the DQ Decoupling feature, find the compile option in config_features.h and verify that it is defined:

#define ENABLE_DQ_DECOUPLING

3.2 Firmware Options

To configure DQ Decoupling, please verify that the desired options are defined in config_features.h and that the selections and values are appropriate for the application:

DQ Decoupling Method Selection

SEL_DQ_DECOUPLING_METHOD

Use to select DQ Decoupling method. The available options are:

FEEDFORWARD_DECOUPLING

FEEDBACK_DECOUPLING

3.3 Firmware Parameters

DQ Decoupling uses the firmware specified configuration parameters below. These parameters are defined in config_features.h:

CURRENT_FILTER_CUTOFF_FREQ_Q16
Maximum cutoff frequency (Hz) for low-pass filter applied to motor current for the decoupling calculation. The filter cutoff frequency is calculated based on the current PI controller bandwidth unless it exceeds this maximum setting.

3.4 Tuning Parameters (GUI)

There are no GUI configurable parameters for DQ Decoupling.

CONTACT INFORMATION

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IMPORTANT NOTICE